THE FUNCTOR FROM MODULES OVER A FIBRE PRODUCT HAS A RIGHT ADJOINT GIVEN BY A FIBRE PRODUCT OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The Situation of a fibre product of rings with a module over it, and the functor displayed before the statement. situation environment at source lines 521-542 and the functor (equation-functor) at source lines 625-632, both before the counted statement. Fixes the data the statement refers to, namely the diagram of rings B = B' x_A A', the module M', the descent datum phi and the functor whose right adjoint is computed.
- The functor that sends a triple (N, M', phi) to its two base changes has a right adjoint, proved in the chapter's module lemma. the author cites this at proof lines 659-660 as Lemma lemma-modules (tag 0D2F), a statement of the same chapter. Supplies the adjointness half of the assertion, which the author does not reprove; only the identity statement is carried out inside the counted span.
- Exactness and flatness-free manipulation of the fibre product of modules, in particular the identification of kernels of the horizontal maps. used at proof lines 661-679, where the kernels I, J, K, L of the two diagrams are introduced and compared. Supplies the objects in terms of which all twelve observations are formulated.

Source reference. Original lemma 07RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9879–9879. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original section 08KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 525–525. Navigation pointer to the Situation that fixes the rings, the modules and the comparison map of the statement; it contains no proof content and the counted argument does not use anything from it beyond the naming of the data.

Source reference. Original equation 08KI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 629–629. Navigation pointer to the displayed functor whose right adjoint the statement describes; quoting the display is a matter of notation, and the identification of the adjoint and the identity statement are carried out inside the counted proof span.

Source reference. Original lemma 0D2F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16380–16380. Navigation pointer to the chapter's module lemma that the author cites for the adjointness half of the statement; the second half, which is the substance of the card, is proved inside the counted span.

Lemma 0.2. *In Situation 08KH the functor (08KI) has a right adjoint, namely the functor*

$$F : (N, M', \varphi) \longmapsto N \times_{\varphi, M} M'$$

where $M = M'/IM'$. Moreover, the composition of F with (08KI) is the identity functor on $Mod_B \times_{Mod_A} Mod_{A'}$. In other words, setting $N' = N \times_{\varphi,M} M'$ we have $N' \otimes_{B'} B = N$ and $N' \otimes_{B'} A' = M'$.

Proof. The adjointness statement follows from the more general Lemma 0D2F. To prove the final assertion, recall that $B' = B \times_A A'$ and $N' = N \times_{\varphi,M} M'$ and extend these equalities to

$$\begin{array}{ccccc} A & \longleftarrow & A' & \longleftarrow & I \\ \uparrow & & \uparrow & & \uparrow \\ B & \longleftarrow & B' & \longleftarrow & J \end{array} \quad \text{and} \quad \begin{array}{ccccc} M & \longleftarrow & M' & \longleftarrow & K \\ \uparrow{\scriptstyle \varphi} & & \uparrow & & \uparrow \\ N & \longleftarrow & N' & \longleftarrow & L \end{array}$$

where I, J, K, L are the kernels of the horizontal maps of the original diagrams. We present the proof as a sequence of observations:

(1) $K = IM'$ (see statement lemma),
(2) $B' \to B$ is surjective with kernel J and $J \to I$ is bijective,
(3) $N' \to N$ is surjective with kernel L and $L \to K$ is bijective,
(4) $JN' \subset L$,
(5) $\mathrm{Im}(N \to M)$ generates M as an A-module (because $N \otimes_B A = M$),
(6) $\mathrm{Im}(N' \to M')$ generates M' as an A'-module (because it holds modulo K and L maps isomorphically to K),
(7) $JN' = L$ (because $L \cong K = IM'$ is generated by images of elements xn' with $x \in I$ and $n' \in N'$ by the previous statement),
(8) $N' \otimes_{B'} B = N$ (because $N = N'/L$, $B = B'/J$, and the previous statement),
(9) there is a map $\gamma : N' \otimes_{B'} A' \to M'$,
(10) γ is surjective (see above),
(11) the kernel of the composition $N' \otimes_{B'} A' \to M' \to M$ is generated by elements $l \otimes 1$ and $n' \otimes x$ with $l \in K$, $n' \in N'$, $x \in I$ (because $M = N \otimes_B A$ by assumption and because $N' \to N$ and $A' \to A$ are surjective with kernels L and I),

(12) any element of $N' \otimes_{B'} A'$ in the submodule generated by the elements $l \otimes 1$ and $n' \otimes x$ with $l \in L$, $n' \in N'$, $x \in I$ can be written as $l \otimes 1$ for some $l \in L$ (because J maps isomorphically to I we see that $n' \otimes x = n'x \otimes 1$ in $N' \otimes_{B'} A'$; similarly $xn' \otimes a' = n' \otimes xa' = n'(xa') \otimes 1$ in $N' \otimes_{B'} A'$ when $n' \in N'$, $x \in J$ and $a' \in A'$; since we have seen that $JN' = L$ this proves the assertion),

(13) the kernel of γ is zero (because by (10) and (11) any element of the kernel is of the form $l \otimes 1$ with $l \in L$ which is mapped to $l \in K \subset M'$ by γ).

This finishes the proof. □